

Spring Linear Programming Problems

- ① Graphical Solution
- ② Simplex Algorithm

Steps

- ① Identify the decision variables
- ② Write the constraints as linear inequalities
- ③ Plot the constraints on a graph to find the feasible region.
- ④ Identify the corners of the feasible region

⑤ Evaluate the objective
fn at each corner

Problem

Maximize

$$Z = 3x + 2y \text{ is subject}$$

to the constraints

$$1) \quad 2x + y \leq 20$$

$$2) \quad 4x - 5y \geq -10$$

$$3) \quad x \geq 0$$

$$4) \quad y \geq 0$$

soln

① $2x + y \leq 20$

$2x + y = 20$

x	0	10
y	20	0

$\{(x, y): (0, 20), (10, 0)\}$

(1,1)

$2x + y \leq 20$

$2(1) + (1) \leq 20$

$2 + 1 \leq 20$

$3 \leq 20$

$2x + 0 = 20$

$2x = 20$

$x = 10$

$x = 10$

2)

$4x - 5y \geq -10$

$4x - 5y = -10$

x	0	-2.5
y	2	0

(1,1)

$4(1) - 5(1) \geq -10$

$4 - 5 \geq -10$

$-1 \geq -10$

$-5y = -10$

$y = 2$

$4x = -10$

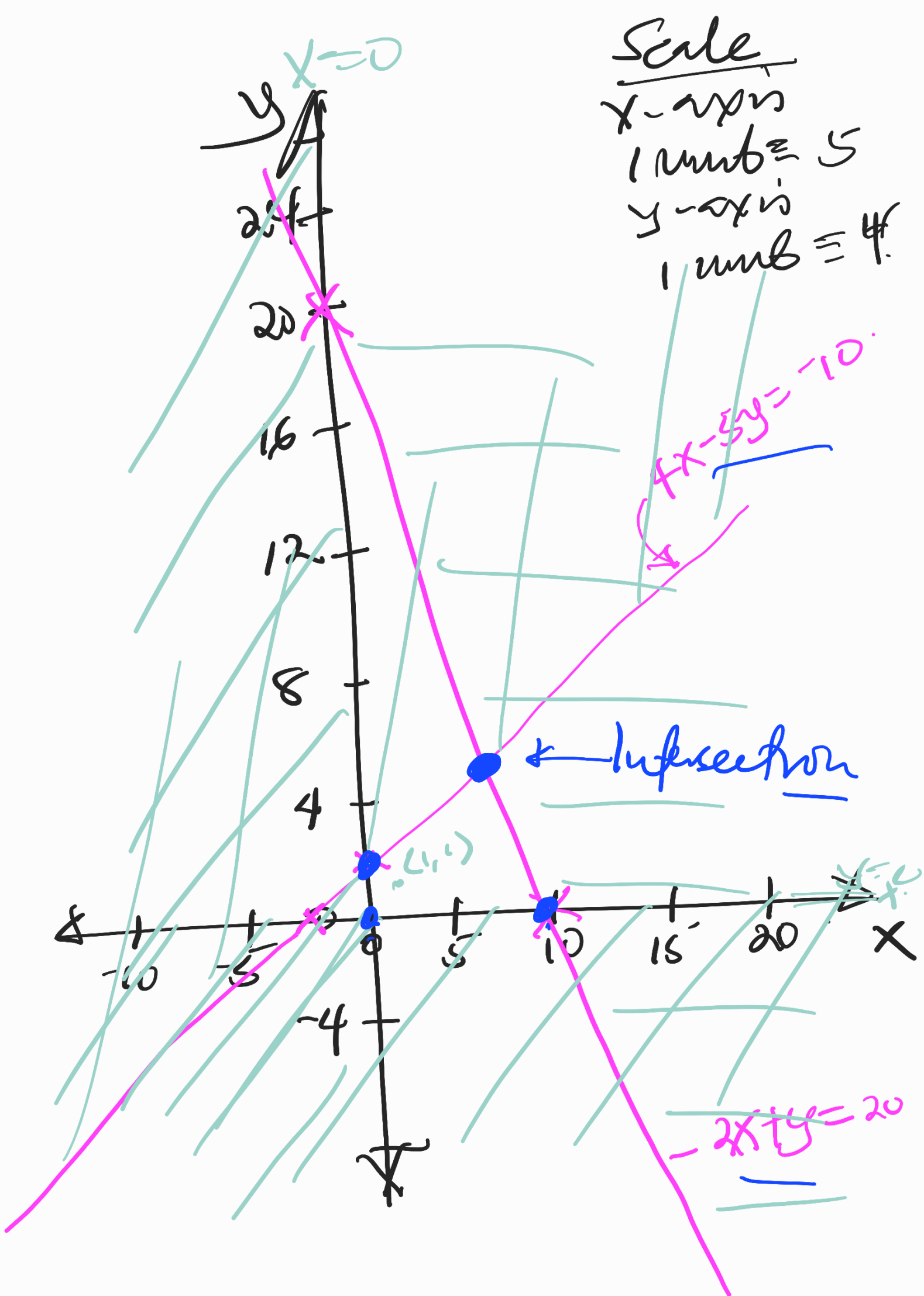
$x = -2.5$

$$\{(x, y) : (0, 2), (-2, 0)\}$$

3) $x \geq 0$ (1, 1)
 $x = 0$ 17, 0 (True)
 y -axis

4) $y \geq 0$
 $y = 0$, x -axis

$y \geq 0$
(1, 1) 17, 0 (True)



Intersection

$$\begin{array}{l} 2 \mid 2x + y = 20 \\ 1 \mid 4x - 5y = -10 \end{array}$$

$$\begin{array}{r} 4x + 2y = 40 \\ -4x - 5y = -10 \end{array}$$

$$-3y = 30$$

$$y = -10$$

$$\underline{y = -10}$$

$$\text{from } 2x + y = 20$$

$$y = 7$$

$$2x + 7 = 20$$

$$2x = 13$$

$$x = 13/2$$

$$x = 6.5$$

$$\{(x, y), (6.5, 7)\}$$

∴ The Intersection
 $\{(x, y): (0, 0), (6.5, 7), (0, 2), (10, 0)\}$

Now maximum

$$Z = 3x + 2y \big|_{(6,0)}$$

$$\underline{\underline{Z = 0}}$$

$$Z = 3x + 2y \big|_{(6.5, 7)}$$

$$Z = 3(6.5) + 2(7)$$

$$Z = 33.5$$

$$Z = 3x + 2y \big|_{(0,2)}$$

$$Z = 3(0) + 2(2)$$

$$\underline{\underline{Z = 2}}$$

$$Z = 3x + 2y \quad | \quad (0,0)$$

$$Z = 3(10) + 2(0)$$

$$Z = \underline{\underline{30}}$$

$$Z = 0, 2, 30, 33.5$$

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